Data bibliography / attribution

Orbital Technosphere / Observer

01 CelesTrak / orbital elements and memberships

26 OMM CSV catalogs. Collected 2026-11-03T17:13:12Z to 2026-11-03T17:14:05Z.
Active catalog supplies the satellite population; membership catalogs identify named fleets. Three bounded debris catalogs supplement it.
Catalogs: active, starlink, oneweb, kuiper, qianfan, hulianwang, gnss, iridium-next, globalstar, orbcomm, resource, science, stations, fengyun-1c-debris, iridium-33-debris, cosmos-2251-debris, planet, spire, intelsat, ses, eutelsat, telesat, gps-ops, glo-ops, galileo, beidou.
Attribution: CelesTrak. Exact feed URLs, element epochs and hashes are retained in the source ledger.
https://celestrak.org/NORAD/documentation/gp-data-formats.php

02 SGP4 / Vallado, Crawford, Hujsak and Kelso

Positions propagated from the pinned elements using the reference SGP4 implementation, WGS-72 and AFSPC mode.
Calculation instant: 2026-11-03T17:10:00Z. 44 records excluded by epoch; 0 propagation failures.
Public element uncertainty and approximate Earth rotation limit these visualization positions.
https://celestrak.org/publications/AIAA/2006-6753/

Historical validation / NASA SSCWeb

Reference instant: 2026-08-13T15:36:00Z.
Historical regression reference only. The November refresh was unreachable; this response does not provide an independent comparison of the plotted November positions.
https://sscweb.gsfc.nasa.gov/WebServices/REST/

Cartography / Cartofreako

Projection: Authagraph. The projection bibliography credits its mathematical and design sources.
Separate map and legend composition follows the Setouchi ST2 layout. Typography: Atkinson Hyperlegible; bundled fonts under the SIL Open Font License.
https://bdekoz.github.io/cartofreako/docs/pages/projections/authagraph/bibliography.html

Pinned inputs / source ledger

The profile records the calculation time, observer, filters and every catalog URL. Adjacent acquisition.json and SHA256SUMS record retrievals and original input bytes.
Map and legend carry the same timestamp and counts; the complete acquisition ledger is embedded in this SVG.
Map image scaled to 98.863636% with proportional sheet margins.
https://github.com/bdekoz/cartofreako/blob/main/assets.static/orbital-technosphere/20261103/orbital-technosphere-profile.json

Setouchi AuthaGraph: +4.35-inch printed horizontal move on a 43.5-inch image. Celestial directions retain source registration in the same cyclic frame.

MSP speculation 3 / Plum / ice blue

Randoma11y: ground #2a0c41; type #73def9. sRGB contrast 11.04:1 (screen only). Vendor markers retain source colors.
https://randoma11y.com/%2373def9/%232a0c41
Epson Stylus Pro 9000 / metallic glossy paper / cyan and light magenta reported low; grays reportedly print violet.
Intended RGB hues; final printed hue is unmeasured. No green correction, ink separation or metallic reflection simulation.

SGP4 positions / 2026-11-03T17:10:00Z

1,088 plotted objects / named fleets and residual roles

Seven-day element age; six-hour future tolerance.

Starlink	549
OneWeb	44
Kuiper	31
Qianfan	20
Hulianwang	21
Iridium	5
Globalstar	4
ORBCOMM	0
Planet	0
Spire	2
Intelsat	38
SES	26
Eutelsat	7
Telesat	12
GPS	11
GLONASS	8
Galileo	13
BeiDou	13
Debris	53
Other active	220
Other navigation	2
Other Earth observation	6
Science	2
Human presence	1

One object per NORAD ID. Overlapping named memberships use profile order.

Name and locpoint sublayers are independent. Names are hidden by default.

Black x: debris. Sunburst: Starlink. Square: Intelsat. Triangle: remaining Earth observation.

Active-marker opacity records illumination. Debris remains solid black.

Topocentric celestial directions above the San Francisco horizon. Equator and horizon are reference lines.

Orbital technosphere Observer / Authagraph